MULTIMEDIA TECHNOLOGIES

BTEUP Polytechnic CSE — Study Guide & Most Important Question Bank

Developer:- Vipin Gangwar

Exactly 30 syllabus-aligned theory questions • Diagrams • Last-Minute Revision

Scope: Covers the four theory units in the supplied syllabus. Practicals are intentionally excluded.

Exam note: These notes support preparation but cannot predict the exact paper. Understand the concepts and adapt answer length to the marks and instructions in your examination.

UNIT 1: INTRODUCTION TO MULTIMEDIA

Q1. Define Multimedia. Explain its core components (Text, Audio, Video, Graphics, Animation) in detail.

Multimedia is the computer-based integration of two or more media types to communicate information. Its components are:
1. Text: conveys titles, instructions, explanations and labels. Font, size, spacing and contrast affect readability.
2. Graphics and still images: photographs, icons, charts, drawings and diagrams explain ideas visually. Raster graphics use pixels; vector graphics use mathematical paths.
3. Audio: speech, music and sound effects add explanation, feedback and atmosphere. Digital audio quality depends on sampling rate, bit depth and compression.
4. Video: a sequence of frames displayed rapidly to create the impression of motion. Resolution, frame rate, codec and bitrate affect quality and file size.
5. Animation: changes in images or object states over time to create movement, demonstrate processes or tell stories.
A multimedia product may also include user interaction, such as buttons, menus and links. For example, an e-learning lesson may combine text, diagrams, narration, video and animated steps. The workflow is to plan content, create or capture each asset, edit it, combine assets in authoring software, test the result and export it in a suitable format.
Multimedia improves engagement and can explain complex concepts, but large files require storage and bandwidth. Good design keeps each media element relevant and accessible.

Q2. Explain the difference between Linear and Non-Linear Multimedia with real-world examples.

Linear multimedia follows a fixed sequence from beginning to end. The audience generally watches or listens without choosing the order. Examples include a cinema film, a television advertisement and a continuously played recorded lecture. Its structure is simple: Start → Scene 1 → Scene 2 → Scene 3 → End. The creator controls sequence and pacing, so linear media is useful when every viewer should receive the same message.
Non-linear multimedia allows users to control navigation and select what to view or do next. Examples include websites, interactive learning modules, video games and digital encyclopaedias. A user might select a menu topic, complete an activity and return to the home screen.
Key differences are sequence (fixed versus selectable), user control (limited versus interactive), and production needs (timing-focused versus navigation and interface design). Non-linear products need clear menus, consistent controls and feedback so users do not become lost. Many products combine both: a game may start with a fixed animation and then provide an interactive environment. Neither type is automatically better; the choice depends on purpose, audience and desired participation.

Q3. Discuss the essential Hardware requirements for a Multimedia System.

A multimedia system needs hardware to capture, process, store, display and transmit media. Essential components include:
1. CPU: runs the operating system and multimedia applications, manages tasks and performs encoding or editing work.
2. RAM: stores active programs, frames and temporary media data. More RAM helps with large images, video timelines and multitasking.
3. GPU: accelerates graphics, 3D rendering, visual effects and supported video operations.
4. Storage: SSDs provide fast loading and editing; high-capacity drives store large media libraries and backups.
5. Input devices: keyboard, mouse, touchscreen, scanner, digital camera, webcam, microphone and graphics tablet capture information.
6. Output devices: monitor, speakers/headphones, printer and projector present media to users.
7. Communication devices: network adapters, Wi-Fi, routers and modems enable file transfer, streaming and conferencing.
8. Supporting hardware: sound interfaces, capture cards, external drives and UPS devices may be needed for specialised work.
Diagram: Input devices → CPU/RAM/GPU ↔ Storage → Monitor/Speakers/Network.
The required specification depends on the task. A slideshow needs less processing power than 4K editing or 3D animation. Compatibility, drivers, cooling, ports and reliable backup are also important.

Q4. Explain the different types of Multimedia Software (Authoring tools, Editing tools, etc.).

Multimedia software creates, edits, combines, compresses, plays and distributes media. Main types are:
1. Authoring tools: combine text, images, audio, video, timing, links, buttons and interaction into a complete product such as a presentation or e-learning module.
2. Image editors: raster editors retouch photographs and pixels; vector editors create scalable logos, diagrams and illustrations.
3. Audio editors: record, cut, mix, filter and export narration, music and sound effects.
4. Video editors: arrange clips on a timeline, trim footage, add titles, transitions, colour correction and sound, then export a finished video.
5. Animation/3D tools: create keyframes, motion paths, models, materials, lighting, rigs and rendered scenes.
6. Desktop publishing tools: arrange text and graphics into brochures, magazines, reports and books.
7. Compression/conversion tools: encode or decode media and convert between formats.
8. Playback/capture tools: play media or capture images, sound, video and screen activity.
Typical workflow: plan → capture/create assets → edit → author/assemble → test → compress/export → distribute.
Tool choice depends on purpose, supported formats, budget, operating system and user skill. Preserve source files and test the exported product on its target device.

Q5. What are the key features required in a Multimedia Operating System? Explain.

A multimedia operating system manages computer resources so media applications run smoothly. Important features include:
1. Predictable scheduling: time-sensitive audio and video tasks need CPU time before playback deadlines.
2. Efficient memory management: large buffers and media assets must be handled without excessive paging or pauses.
3. Fast input/output: storage, cameras, microphones, displays and audio devices need timely data transfer.
4. Device management: drivers coordinate cameras, scanners, sound cards, GPUs, displays and printers.
5. File-system support: large files must be stored, retrieved and protected efficiently.
6. Networking: protocols and buffering support streaming and online communication under changing network conditions.
7. Multitasking: audio may play while other applications download or edit files.
8. Security and reliability: permissions, isolation, updates and backups protect content.
9. User interface/accessibility: consistent controls, keyboard access and captions improve usability.
Diagram: Multimedia app → OS scheduler/memory/I/O services → drivers → hardware.
Applications and codecs perform much of the actual media encoding and decoding; the OS provides resource management and device access. Performance depends on the combined quality of hardware, drivers, software and network services.

Q6. Explain the concept of a Multimedia Communication System and data transmission.

A multimedia communication system captures, encodes, transmits, receives and presents text, images, audio or video between locations. Examples include video calls, online classes and live streaming.
Main components: source (camera/microphone/computer), digitiser, encoder/compressor, packetisation and transport, network, receiver/decoder, and output devices.
Diagram: Camera/Microphone → Digitisation → Compression/Encoding → Packets → Network → Buffer/Decode → Screen/Speakers.
Analogue sound is sampled and quantised to become digital data. The encoder may compress it to reduce bitrate. Data is divided into packets and transmitted using network protocols. The receiver collects packets, handles ordering or loss, decodes the stream and presents it.
Transmission can be simplex (one direction), half-duplex (both directions but not simultaneously) or full-duplex (both simultaneously). Multimedia quality depends on bandwidth, latency, jitter and packet loss. Buffering smooths arrival variations but may increase delay. Error correction, adaptive bitrate and congestion control improve delivery. Real-time conversation prioritises low latency, while file transfer can often wait for retransmission. Synchronisation, privacy, security and codec compatibility are also important.

Q7. Discuss the role of bandwidth and streaming in multimedia communication.

Bandwidth is the capacity of a network link, commonly measured in bits per second. Streaming delivers media so playback begins before the entire file has downloaded.
Workflow: Media source → encoder/streaming server → network → player buffer → decoder → screen/speakers.
If the media bitrate is higher than the sustained network throughput, the buffer may empty and playback pauses.
Resolution, frame rate, scene complexity and audio quality influence required bitrate. Advertised bandwidth may differ from actual throughput because of congestion, Wi-Fi interference or competing users.
The player stores a short amount of incoming data in a buffer to smooth network variation. A larger buffer reduces interruptions but can increase start-up time and live delay. Adaptive bitrate streaming offers several encodings and selects a suitable quality based on estimated network conditions. It can lower quality when bandwidth drops and raise it when conditions improve.
Key terms: bitrate is encoded data per second; throughput is actual successful delivery rate; latency is travel delay; jitter is variation in delay; CDN means content delivery network. Compression reduces bandwidth needs but excessive compression damages quality. Live calls prioritise low delay, while on-demand video can use more buffering.

UNIT 2: BASIC COMPRESSION TECHNIQUES

Q8. What is Data Compression? Why is it necessary for Multimedia? Explain Lossy vs Lossless compression.

Data compression represents information with fewer bits. It reduces file size, storage needs, transmission time and bandwidth consumption—important because raw audio, images and video can be very large.
Lossless compression reconstructs the original data exactly. It exploits repeated patterns, statistical probabilities or predictable relationships. Examples include Huffman coding, RLE, LZMA, FLAC and ALAC. It is appropriate for text, program files, archives and audio that must retain exact samples.
Lossy compression permanently discards some information to obtain greater size reduction. It may exploit limits of human hearing or vision. Examples include JPEG for photographs, MP3 for audio and common video coding systems. It is useful for streaming and online media but may create blur, blocking or audible artefacts.
Flow: Original → Encoder/compressor → Compressed data → Decoder → Reconstructed data.
Compression ratio = original size ÷ compressed size. If a 10 MB file becomes 2 MB, the ratio is 5:1 and the size reduction is 80%. Results depend on content. Lossless is preferred when exact fidelity matters; lossy is often preferred when small files and efficient distribution matter. Quality, compatibility, processing time and latency must also be considered.

Q9. Explain the Huffman Coding algorithm step-by-step with a logical example and a tree diagram.

Huffman coding is a lossless entropy-coding method that assigns shorter binary codes to frequent symbols and longer codes to rare symbols.
Steps: (1) count symbol frequencies; (2) create one leaf per symbol; (3) select the two lowest-frequency nodes; (4) combine them into a parent whose frequency is their sum; (5) repeat until one root remains; (6) label branches 0 and 1; (7) read each root-to-leaf path as the symbol's code.
Example frequencies: A=5, B=2, C=1, D=1. Combine C(1)+D(1)→CD(2); combine B(2)+CD(2)→node(4); combine node(4)+A(5)→root(9).
Tree:
(9)
/ \
0 / \ 1
(4) A(5)
/ \
0 / \ 1
B(2) (2)
/ \
0 / \ 1
C(1) D(1)
Codes: A=1, B=00, C=010, D=011. Ties can produce different but valid trees. For message ABACD, encoded bits are 1|00|1|010|011. Huffman codes are prefix-free, so no complete code is the prefix of another; decoding is unambiguous by following the tree until a leaf is reached. It is often combined with other compression stages.

Q10. Explain Run Length Encoding (RLE) with clear examples. Where is it most effective?

Run Length Encoding (RLE) is a lossless method that stores a consecutive sequence of identical values as a value plus a count.
Steps: scan input from left to right, count repeated values, output each value-count pair, and start a new run when the value changes.
Example: AAAABBBCCDAA → (A,4)(B,3)(C,2)(D,1)(A,2).
Binary row: 0000000011111000 → (0,8)(1,5)(0,3).
The decoder repeats each value according to its count, reconstructing the original sequence exactly.
RLE is effective for simple icons, line drawings, monochrome images, masks, diagrams and documents containing long runs of the same pixel. It is simple, fast and uses little processing power. It may perform poorly on photographs or random data because runs are short; storing counts may then make the file larger. A real format needs rules for storing counts and distinguishing run markers from literal data.
Diagram: Input A A A A B B C C C → count runs → (A,4)(B,2)(C,3) → decode → original sequence.
The key principle is exploiting repetition. Compression effectiveness depends on the structure of the input rather than merely its size.

Q11. Explain the working principle and logic of the JPEG standard for image compression.

JPEG is a common image-compression standard, especially for photographs. Typical baseline JPEG is lossy. Its stages are:
1. Colour conversion: RGB is commonly converted to YCbCr, separating brightness (Y) from colour-difference components (Cb and Cr).
2. Chroma subsampling: colour information may be stored at lower resolution because vision is generally more sensitive to brightness detail.
3. Block division: components are divided into 8×8 pixel blocks.
4. Discrete Cosine Transform (DCT): converts pixel values into frequency coefficients. Low frequencies represent broad changes; high frequencies represent fine detail.
5. Quantisation: coefficients are divided by quantisation values and rounded. This is the main lossy stage.
6. Ordering and run handling: coefficients are arranged so many zero values can be coded efficiently.
7. Entropy coding: differential, run-length and Huffman coding reduce redundancy.
Flow: RGB → YCbCr → subsampling → 8×8 blocks → DCT → quantisation → entropy coding → JPEG.
Higher quality settings usually create larger files. Strong compression can cause blocking, ringing and blurred detail.
JPEG works well for photographs but is less suitable for sharp text and line art. The DCT itself need not lose information; quantisation makes common JPEG lossy. Decoding reverses the stages approximately to reconstruct the image.

Q12. Explain the MPEG standard for video compression. Detail the logic of I-frames, P-frames, and B-frames.

MPEG is a family of standards for compressing digital audio and video. Video compression uses spatial redundancy within frames and temporal redundancy between neighbouring frames. The encoder may predict motion, encode the difference from a prediction, transform and quantise residual data, then apply entropy coding.
I-frame: intra-coded independently of other frames. It is useful for seeking and recovery but is often larger.
P-frame: predicted from a previous reference frame, using motion vectors and residual differences; generally smaller than an I-frame.
B-frame: may use both earlier and later reference frames for bidirectional prediction, often improving compression. Frame reordering may add delay.
Conceptual sequence: Display order I → B → B → P → B → P; B-frames may reference earlier and later I/P frames.
Flow: Frames → motion estimation/prediction → residual → transform/quantisation → entropy coding → bitstream.
The decoder reconstructs reference frames and uses motion vectors to rebuild predicted frames. Quantisation is lossy, while prediction and entropy coding need not be. MPEG versions differ, so MPEG is a family rather than one codec.
Benefits include lower storage and bandwidth; limitations include encoding complexity, artefacts at low bitrates and possible delay. In an exam, explain frame roles and motion vectors clearly.

Q13. Compare MP3 and MP4 audio compression standards with their logic.

MP3 generally refers to MPEG Audio Layer III, a lossy audio-coding format. MP4 usually refers to an MPEG-4 multimedia container, not one specific audio codec. An MP4 file may contain AAC audio, H.264/H.265 video, subtitles and metadata.
MP3 logic: the encoder analyses audio in short segments, transforms it into frequency-related information, uses a psychoacoustic model to identify less audible components, quantises information and codes it efficiently. The decoder reconstructs an approximation of the original sound. Lower bitrates reduce file size but may cause audible artefacts.
MP4 logic: the container organises encoded streams into tracks and stores timing, indexing and metadata so audio and video can be synchronised and searched. Its quality depends on the codecs and settings inside it.
Diagram: Audio → MP3 encoder → .mp3 → decoder → sound. Audio/video → codecs → MP4 container → player separates tracks → decoders → output.
Comparison: MP3 is primarily an audio format/codec; MP4 is a container. MP3 typically stores compressed audio; MP4 can carry multiple media streams. Neither name alone guarantees quality—source quality, codec and bitrate matter. Avoid saying MP4 is simply an audio codec that replaced MP3.

Q14. Give a brief logical overview of LZMA and FLAC.

LZMA (Lempel–Ziv–Markov chain Algorithm) is a general-purpose lossless compression method used in archive formats such as 7z. It finds repeated sequences and represents later occurrences using references to earlier data, then uses probability modelling and range coding. A large dictionary can achieve high compression on text, software and repetitive files, but may require considerable memory and encoding time.
Flow: Input → find repeated strings/dictionary matches → probability modelling → range coding → compressed stream. The decoder reconstructs the original bytes exactly.
FLAC (Free Lossless Audio Codec) is designed for digital audio. It predicts samples from neighbouring samples, calculates residual differences and codes them efficiently. The decoder reverses the process to reproduce the original PCM samples exactly.
Flow: PCM audio → prediction → residual coding → FLAC file → decoding → original PCM.
Both are lossless, but their targets differ: LZMA is general-purpose data compression, while FLAC is audio-specific. LZMA suits archives and software distribution; FLAC suits music libraries and preservation. Neither guarantees a fixed compression ratio; already-compressed data may shrink very little.

Q15. Give a brief logical overview of ALAC and ITU G.722.

ALAC (Apple Lossless Audio Codec) compresses digital audio while preserving the original decoded sample values. It predicts samples, calculates residuals and encodes the residual information. Its decoder reconstructs the original PCM samples. ALAC is useful for music libraries and audio preservation where fidelity matters.
Flow: PCM samples → prediction → residual coding → ALAC stream → decode → original samples.
ITU-T G.722 is a wideband speech-coding standard used in voice communication. It uses sub-band adaptive differential pulse-code modulation (SB-ADPCM). Audio is split into frequency sub-bands and each band is coded using adaptive prediction/difference techniques. It is designed for natural-sounding speech at relatively modest bitrates and is used in IP telephony and conferencing.
Comparison: ALAC is lossless and targets faithful storage/playback; G.722 is a lossy speech codec designed for real-time communication. ALAC suits archives and music; G.722 prioritises intelligibility, bandwidth and low delay. The correct codec depends on application requirements rather than assuming one is universally better.

Q16. Explain the logic of H.261 and H.265 video coding standards.

H.261 and H.265 are ITU-T video-coding standards from different generations. Both exploit spatial redundancy within frames and temporal redundancy between frames.
H.261 was designed for early video telephony and conferencing. It uses macroblocks, motion-compensated prediction, transform coding, quantisation and variable-length coding. The encoder predicts a frame from a reference, calculates residual differences, transforms and quantises them, then codes the result.
Flow: Frame → macroblocks → motion prediction → residual → transform/quantisation → coding → bitstream.
H.265, also called HEVC, aims for higher compression efficiency, especially for high-resolution video. It uses more flexible coding-tree units, improved intra/inter prediction, motion tools, transforms, filters and entropy coding. This can provide similar visual quality at a lower bitrate than older methods, although encoding is more complex and device support matters.
Comparison: H.261 is an early conferencing standard; H.265 is a modern high-efficiency codec. H.261 uses simpler macroblock structures, while H.265 supports flexible partitions and advanced prediction. Both are commonly lossy because quantisation discards information. H.265 is used for UHD video, streaming and storage; H.261 is mainly historically important. Explain that the improvement comes from better prediction and coding tools, not simply from supporting higher resolution.

UNIT 3: CONTENT DEVELOPMENT AND DISTRIBUTION

Q17. Define Desktop Publishing (DTP). Explain its workflow in multimedia.

Desktop Publishing is the use of computers and page-layout software to combine text, photographs, illustrations and other elements into professional print or digital pages. Typical products include brochures, magazines, posters, reports, catalogues and books.
Workflow: (1) plan purpose, audience, size and output; (2) collect text, images, logos and charts; (3) set page size, margins, columns, grid and styles; (4) place text and images in frames; (5) refine typography, alignment, contrast, spacing and hierarchy; (6) proofread spelling, page numbers, clipping and image quality; (7) preflight fonts, links, bleed and colour settings; (8) export a print-ready PDF or digital file and distribute it.
DTP differs from word processing because it offers precise control of page composition, text flow and graphic placement. In multimedia, assets may be created in graphics editors and then assembled in the publication. Advantages include consistent layouts, reusable templates and easy revision. Limitations include the need for design skill and careful font/image management. Good DTP is both attractive and readable; software alone cannot replace clear content and sound design principles.

Q18. Compare the roles of CorelDRAW, Photoshop, and PageMaker in content creation.

CorelDRAW is primarily a vector-graphics application. It creates shapes using points, lines and mathematical paths, making it suitable for logos, icons, diagrams, signage and scalable illustrations.
Adobe Photoshop is primarily a raster-image editor. It works with pixels and is used for photograph retouching, colour correction, masking, compositing, digital painting and special effects. Raster images have fixed pixel dimensions, so excessive enlargement may reduce sharpness.
Adobe PageMaker was a desktop-publishing application for arranging text and graphics in multi-page publications. It supported page layout, columns and text flow. It is legacy software and has largely been replaced by newer publishing tools.
Comparison: CorelDRAW = vector drawing; Photoshop = pixel/image editing; PageMaker = page layout. A brochure workflow might create a logo in CorelDRAW, retouch photos in Photoshop, then assemble text and images in a DTP application. Vector graphics scale cleanly; raster quality depends on resolution; layout software manages composition across pages. Choose tools according to the asset and output requirements. Compatible formats and colour profiles help maintain quality when moving work between applications.

Q19. Explain the concept of Multimedia Animation.

Multimedia animation creates the illusion of movement by displaying a sequence of images or changing object states over time. It is used in films, games, education, advertising, simulations and interfaces. A frame is one image/state; frame rate is the number displayed per second.
Production stages: (1) concept and storyboard; (2) design characters, objects and backgrounds; (3) create 2D assets or 3D models; (4) define key poses, paths and timing; (5) add lighting and effects; (6) render and composite layers; (7) edit with sound and narration; (8) export and test on target devices.
Types include hand-drawn, 2D digital, 3D computer animation, stop motion and motion graphics. Pre-rendered animation is delivered as a video; real-time animation is generated during interaction, such as in games.
Animation can make complex or invisible processes easier to understand and attract attention. However, it may be expensive or distracting if used without purpose. Good animation uses consistent design, believable timing, clear movement and appropriate accessibility support such as captions. Animation should strengthen the message rather than merely decorate the screen.

Q20. Detail the core logical differences between 2D and 3D Animation.

2D animation mainly works in horizontal X and vertical Y dimensions. Assets are drawings, vector shapes, sprites or layers. Motion is created by changing drawings, positions, rotations, shapes or layer properties. It is common in cartoons, explainers and motion graphics.
3D animation represents objects in X, Y and Z space. Artists create models, materials, textures, lights and cameras, then animate objects or character rigs. A virtual camera projects the 3D scene into a 2D image.
Differences: 2D assets are generally flat; 3D models have depth. 2D can be efficient for stylised graphics, while 3D supports camera movement and different viewpoints but often needs more modelling and rendering resources. In 2D, new poses or layer movement create motion; in 3D, objects and rigged joints move within a scene.
Pipelines: 2D—storyboard → drawings/assets → keyframes/in-betweens → compositing → export. 3D—storyboard → modelling → materials/rigging → animation → lighting/camera → rendering → compositing.
Neither is inherently superior. A simple logo animation may suit 2D; a product needing rotating views may suit 3D. Productions can combine both.

Q21. Explain Keyframes and Tweening with examples.

A keyframe defines an important state of an object at a particular time. It may specify position, rotation, scale, opacity, colour or shape. Tweening (in-betweening) generates intermediate states between keyframes to create smooth motion.
Example: at frame 1, a ball is at x=0; at frame 11, it is at x=100. A linear tween creates intermediate positions at regular intervals. Easing changes the rate of movement so an object may accelerate or decelerate naturally. A bounce requires additional keyframes for rising, peak and falling positions.
Types include motion tweening (position/rotation/scale), shape tweening (morphing compatible shapes), path animation (movement along a curve) and easing (adjusting timing).
Workflow: place object → create first keyframe → move playhead → change property → create next keyframe → preview → refine timing/easing → export.
Keyframes give the animator control over important poses; tweening reduces the need to create every frame manually. Automatic interpolation cannot always produce convincing character movement, so extra keyframes or manual correction may be required. In a diagram, draw a timeline with keyframes at the beginning and end and intermediate frames between them.

Q22. Discuss the historical and technical role of Adobe Flash in animation.

Adobe Flash grew from FutureSplash and Macromedia Flash before Adobe acquired Macromedia. It became influential for web animation, browser games, advertisements and interactive learning content.
Flash combined vector graphics, a timeline, keyframes, motion/shape tweening, audio/video support and ActionScript for interaction. Designers arranged layers and frames, created animated transitions, programmed buttons or games, tested the project and published it commonly as a SWF file played by Flash Player.
Its vector-based workflow could keep many animations relatively compact and made rich interactive content widely available when web standards were less capable. It influenced modern timeline-based animation and interactive design.
Limitations included security and maintenance concerns, inconsistent mobile support, accessibility issues and dependence on a proprietary plug-in. As HTML5, CSS, JavaScript, SVG, canvas and modern video matured, the web moved away from Flash. Adobe ended Flash Player support at the end of 2020, and major browsers removed or blocked its runtime.
Today, Flash is mainly of historical interest; legacy content may need migration or preservation. Modern projects usually use web standards or dedicated animation tools. Important concepts that remain relevant are timelines, keyframes, tweening, vector animation and event-driven interaction.

Q23. Explain Special Effects (VFX) and how they are integrated into modern multimedia.

Special effects create or enhance events that are difficult, costly or impossible to capture directly. Visual effects (VFX) often combine filmed footage with digital imagery. Practical effects, by contrast, are produced physically on set. Techniques include compositing (combining layers), chroma keying (replacing a green/blue background), CGI (computer-generated objects or scenes), motion tracking, masking/rotoscoping, colour grading and simulations of smoke, fire, water or particles.

Workflow: plan → capture footage → create assets → track camera/object movement → mask or key subjects → composite layers → match lighting/colour → review → render final output.

Example: an actor is filmed in front of a green screen. The artist removes the green, tracks camera movement, adds a 3D city, matches perspective and lighting, and composites shadows and colour so the elements appear in one scene.

VFX is used in films, games, advertisements, education, product demonstrations and virtual production. It enables imaginative storytelling and clear visualisation. Challenges include cost, rendering time and making elements look consistent. Perspective, lighting, motion blur, grain, colour and occlusion must match. Poor effects distract from the message. Ethical care is needed when edited visuals could mislead viewers, especially in news or documentary contexts.

UNIT 4: INTRODUCTION TO DIGITAL IMAGING

Q24. Explain the Basics of Graphic Design and the use of Digital Technology.

Graphic design arranges text, images, shapes, colour and space to communicate a message to a specific audience. Its principles include contrast (emphasis), alignment (order), repetition (consistency), proximity (relationships), hierarchy (reading order), balance and whitespace. Typography and colour influence tone and legibility.
Digital technology provides raster editors, vector tools, page-layout software, cameras, scanners, drawing tablets and colour-managed displays. Layers, masks, undo, templates and non-destructive editing make revision easier.
Workflow: brief/research → sketch → create assets → choose typography/colour → arrange composition → obtain feedback → revise → proof/export → distribute.
Screen graphics generally use RGB and suitable pixel dimensions. Print workflows may require CMYK conversion, sufficient effective resolution and bleed. Export settings affect file size, quality and compatibility.
Digital tools enable rapid editing, duplication and distribution, but software alone does not guarantee effective design. Too many fonts, poor contrast, clutter and irrelevant effects reduce clarity. Designers should consider audience, purpose, accessibility, copyright and the output medium. A good poster, for example, makes its main message, date, venue and call to action easy to find quickly.

Q25. Define Digital Image. Explain the concepts of Pixels and Resolution (DPI/PPI).

A digital image stores visual information as numerical data. A raster image is a grid of pixels, each holding colour or intensity information. A pixel is a picture element and the smallest addressable unit of a raster image.
Dimensions state the number of pixels horizontally and vertically. A 1920×1080 image contains 2,073,600 pixels, approximately 2.07 megapixels. Dimensions alone do not determine physical print size.
PPI means pixels per inch and describes how many image pixels are placed in each inch of output. Print width in inches = pixel width ÷ PPI. For example, a 3000-pixel-wide image at 300 PPI prints 10 inches wide.
DPI means dots per inch and technically describes printer ink/toner dots. A printer may use multiple dots to reproduce one image pixel, so DPI and PPI are not identical, although the terms are often confused.
Sharpness also depends on focus, lens quality, compression, colour depth, display and viewing distance. Enlarging a low-resolution image cannot recover detail that was never captured. Vector images scale more cleanly because they are described mathematically. For an exam, draw a grid and label one square “pixel,” then show that the same pixel count at a higher PPI occupies a smaller physical size.

Q26. What is the difference between Raster and Vector graphics? Explain with examples.

Raster graphics store colour values in a grid of pixels. Photographs, scanned pictures and digital paintings are usually raster. Common formats include JPEG, PNG, GIF and TIFF. Raster graphics represent complex textures and colour transitions well, but have fixed dimensions and may blur or pixelate when enlarged excessively.
Vector graphics store shapes as mathematical paths, points, curves, fills and strokes. Logos, icons, diagrams, maps and many illustrations are often vector. Common formats include SVG, EPS and PDF when vector content is retained. Vector objects can scale cleanly because they are rendered at the required size.
Comparison: representation—pixels versus paths; scaling—resolution-dependent versus generally resolution-independent; best use—photos versus clean shapes; editing—pixel-level adjustments versus object/path editing; file size—depends on pixel dimensions and compression versus object complexity.
Diagram:
Raster: [pixel][pixel][pixel]
[pixel][pixel][pixel]
Vector: Point A •■■■■ curved path ■■■■• Point B
A practical workflow may create a logo as vector artwork, retouch product photographs as raster images, and place both in a publication. Exporting vector artwork to a raster format converts it to pixels. Neither type is always better: choose according to content and output.

Q27. Explain the role of digital imaging in modern multimedia applications.

Digital imaging is the capture, creation, processing, storage and display of images in digital form. It supports communication, analysis and visual storytelling.
Applications include: education (diagrams and annotated lessons); entertainment (textures, animation frames and VFX); websites/mobile apps (icons, product images and interface graphics); medicine/science (medical images and microscopy); business (catalogues and advertising); mapping/engineering (satellite images and technical drawings); and social media (captured, edited and shared photos).
Pipeline: capture/create → digitise → process/edit → compress/store → transmit/publish → display/print.
Images may come from cameras, scanners, drawing tools or 3D software. Processing includes cropping, resizing, colour correction, sharpening, noise reduction and compositing. Compression reduces storage and network requirements. Metadata can store capture details or colour information.
Benefits include rapid editing, easy duplication, remote sharing and integration with audio/video. Limitations include resolution constraints, compression artefacts, privacy risks and the possibility of misleading manipulation. Optimisation matters: very large files slow websites and consume mobile data, while excessive compression destroys detail. Use appropriate dimensions, supported formats, alternative text for accessibility, secure storage and proper permissions.
Digital imaging includes any visual asset in a multimedia product, not only photographs.

Q28. How do color depth and color model (RGB vs CMYK) affect digital imaging?

Colour depth is the number of bits used to represent a pixel's colour. More bits allow more possible values and smoother gradients but may require more storage. One-bit imagery represents two states; 8-bit grayscale provides 256 levels; 24-bit RGB commonly uses 8 bits per channel and represents about 16.7 million colours. Some 32-bit formats include an 8-bit alpha/transparency channel.
RGB is an additive colour model used by displays and many digital cameras. Red, green and blue light combine; all channels at maximum produce white, while all at zero produce black.
CMYK is a subtractive model used in many print workflows. Cyan, magenta and yellow inks absorb portions of light, while black ink improves dark tones and printing efficiency.
RGB generally supports bright colours suitable for screens; many CMYK print conditions have a smaller colour gamut. Converting between them can change colours. Colour profiles and soft proofs help predict output, and the printer's requirements should guide conversion.
Diagram: RGB—red + green + blue light → white. CMYK—cyan, magenta, yellow and black inks on paper.
Colour depth affects banding and editing flexibility; colour model affects how devices reproduce colours. Monitor calibration, paper, ink and lighting also affect results. Remember: RGB for light-emitting screens, CMYK for many print processes, with colour management used to reduce unexpected shifts.

Q29. Explain the basic steps involved in digital image processing.

Digital image processing uses algorithms to improve, transform or analyse digital images.
1. Acquisition: capture with a camera/scanner or load an image file.
2. Pre-processing: crop, resize, correct distortion or reduce noise.
3. Enhancement: adjust brightness, contrast, colour, sharpening or smoothing to improve appearance.
4. Restoration: attempt to reduce known degradation such as blur using a model of the degradation.
5. Geometric/colour processing: rotate, scale, align, warp or convert colour spaces.
6. Segmentation: divide an image into meaningful regions or objects.
7. Feature extraction/analysis: measure edges, shape, texture, colour or area; recognition systems may classify objects.
8. Compression/storage: save the result in a suitable lossless or lossy format.
9. Output/evaluation: display, print, transmit or use the image, then check quality and accuracy.
Flow: Acquire → Pre-process → Enhance/Restore → Transform/Segment → Analyse → Store/Display → Evaluate.
Not every application uses every stage. Photo editing may focus on enhancement; scientific analysis may require validated segmentation and measurements. Filtering, thresholding, edge detection and histogram equalisation are common techniques. Processing can introduce artefacts, so preserve originals when possible. In high-stakes contexts, an attractive image is not necessarily an accurate one; results must be validated.

Q30. Discuss the advantages and limitations of digital imaging over traditional analog imaging.

Digital imaging stores visual data electronically, while analogue imaging records a continuous physical or chemical representation, such as a film photograph.
Advantages: immediate review of capture; easy editing of colour, contrast and crop; fast duplication and sharing; efficient indexing and storage; integration with audio, video and interactive media; flexible output for phones, websites and print; and automation of repetitive tasks.
Limitations: files can be deleted or corrupted, so backups are essential; low-resolution images cannot reveal missing detail; lossy compression creates artefacts; devices and electricity are required; proprietary formats can become obsolete; and digital manipulation can raise authenticity concerns. Metadata and image libraries can also create privacy/security risks.
Analogue imaging may offer distinctive aesthetic qualities and a physical original, but it may need chemical processing, careful storage and specialised duplication. Digital systems are convenient but require disciplined file management. Good practice includes retaining originals, using non-destructive edits, maintaining multiple backups, documenting significant changes, choosing suitable resolution/compression, and checking consent and licences. Digital is not automatically superior in every situation. The best method depends on quality, cost, workflow, authenticity and preservation needs.

LAST-MINUTE REVISION

Revise these high-value keywords. For each, be ready to define it, explain its purpose and give an example.

Multimedia: Integration of text, graphics, audio, video and animation.

Linear / non-linear: Fixed sequence versus user-controlled navigation.

Huffman: Frequency-based variable-length prefix codes; merge the two least frequent nodes.

RLE: Value-count pairs; effective for long runs of repeated values.

JPEG: YCbCr, chroma subsampling, 8×8 DCT, quantisation, entropy coding.

MPEG: I-frame = intra; P-frame = predicted; B-frame = bidirectional prediction.

MP3 / MP4: MP3 is an audio coding format; MP4 is a multimedia container.

LZMA / FLAC / ALAC: LZMA general lossless compression; FLAC and ALAC lossless audio.

G.722: Wideband speech coding using sub-band ADPCM.

H.261 / H.265: Video coding standards; H.265/HEVC improves coding efficiency.

DTP: Page layout, typography, image placement, proofing and export.

CorelDRAW / Photoshop / PageMaker: Vector drawing / raster editing / legacy desktop publishing.

Keyframe / tweening: Important state / generated intermediate states.

VFX: Compositing, chroma key, CGI, tracking, masking and colour matching.

Pixel / PPI / DPI: Image sample / pixels per inch / printer dots per inch.

Raster / vector: Pixel grid versus mathematical paths.

RGB / CMYK: Additive screen light versus subtractive print inks.

Image processing: Acquisition, pre-processing, enhancement, segmentation, analysis and output.

EXAM PRESENTATION CHECKLIST

• Start with a precise definition and explain working steps in the correct order.

• Use labelled diagrams for Huffman coding, JPEG workflow, MPEG frame prediction, communication systems and raster/vector graphics.

• Expand technical abbreviations at first use: DCT, DTP, PPI, DPI, VFX, HEVC and ADPCM.

• For comparisons, use paired points and clearly distinguish the concepts.

• For compression questions, state whether the method is lossy or lossless.

• Use accurate examples and avoid irrelevant filler.

Help & Support: t.me/axv_god

Pre-caution: These notes are strictly for your best exam preparation. These notes do NOT guarantee passing the exam. Please do not live in this illusion and do not rely solely on this; study thoroughly and aim to gain excellent marks.